

SDG selected- Climate Action

SDG focuses on

SDG 13 focuses on **taking urgent action to combat climate change and its impacts**. The goal recognizes that rising global temperatures, extreme weather events, sea-level rise, and other climate-related changes threaten people, ecosystems, and economies worldwide.

Key areas of focus include:

1. **Strengthening resilience and adaptation**
 - Help communities prepare for and cope with climate-related disasters such as floods, droughts, storms, and heat waves.
2. **Integrating climate measures into policies**
 - Encourage governments to include climate considerations in national planning, development strategies, and decision-making.
3. **Improving education and awareness**
 - Increase public understanding of climate change, its impacts, and ways to reduce risks.
4. **Supporting developing countries**
 - Mobilize financial resources, technology, and capacity-building to help vulnerable countries address climate challenges.
5. **Implementing international climate agreements**

Advance global cooperation through agreements such as the Paris Agreement, which aims to limit global warming and strengthen climate resilience.

Why this issue is important?

Climate action is important because climate change affects nearly every aspect of human life and the environment. Without effective action, its impacts can become more severe and costly.

Some key reasons why this issue is important:

- **Protecting human lives:** Extreme weather events such as heat waves, floods, droughts, and storms can cause injuries, deaths, and displacement of communities.

- **Food security:** Changes in temperature and rainfall can reduce crop yields, threaten agriculture, and increase the risk of hunger.
- **Water resources:** Climate change can lead to water shortages in some regions and increased flooding in others.
- **Human health:** Rising temperatures and changing environmental conditions can increase heat-related illnesses, spread certain diseases, and worsen air quality.
- **Economic stability:** Damage to infrastructure, agriculture, and businesses can result in significant economic losses.
- **Protecting ecosystems and biodiversity:** Many plant and animal species struggle to adapt to rapid climate changes, increasing the risk of extinction.
- **Future generations:** Actions taken today influence the quality of life and environmental conditions that future generations will inherit.

Problem Statement

Global climate change is one of the most critical challenges of our time, directly targeted by UN Sustainable Development Goal 13 (Climate Action). While international treaties outline carbon-reduction targets, individuals lack clean, visual, and actionable feedback loops to evaluate how their daily energy, travel, and food habits accumulate into environmental impacts.

Current carbon calculators are static, dry worksheets that do not suggest actionable changes. Furthermore, the integration of green energy grids faces grid demand balancing issues, which require intelligent edge load management. There is a lack of educational tools demonstrating how artificial intelligence and neural networks can actively manage clean energy transitions in smart homes.

Proposed Solution

The proposed solution, EcoTrace AI, is an interactive, browser-based prototype designed to address SDG 13 (Climate Action). It translates macro-level sustainability goals into concrete, actionable steps for individuals by combining carbon analytics with client-side Artificial Intelligence.

The solution is divided into five core interactive components:

1. Interactive Carbon Footprint Calculator

Users enter metrics on home utilities, daily travel, flights, dietary habits, and recycling. The `ai-engine.js` calculates live carbon emissions (CO₂e per year) across three key sectors:

Energy: Grid power, heating fuel, and solar offsets.

Transport: Weekly commute mileage, vehicle fuel types (Gas, Hybrid, EV), and flight hours.

Lifestyle & Diet: Meat consumption frequency, food waste levels, and sorting.

2. AI Recommendation Engine

Using rules written in

`ai-engine.js`

, the app evaluates the user's highest emission sources and dynamically generates three prioritized, actionable mitigation tips (e.g., transition to EV, adjust thermostat, reduce food waste) showing exact tons of potential carbon savings.

3. "EcoBot" AI Conversational Assistant

A client-side natural language processing (NLP) chat interface in `app.js`

. Users can type questions about climate terms, carbon offsets, or click suggested prompt buttons to receive instant explanations.

4. AI Smart Grid Neural Network Simulator

A custom Artificial Neural Network (ANN) with a backpropagation learning cycle built from scratch in JS. When the user clicks "Start Training", the canvas visualizes neurons adjusting synaptic weights in real-time as the network learns to manage household battery reserves and dump peak loads, illustrating how AI can prevent utility grid stress.

5. Eco Goals & Impact Tracker

A gamified checklist. Toggling goals (like switching to LED lighting or taking public transit) calculates their exact carbon offset (in kg of CO₂), updates the breakdown chart progress dashboard, and animates the SVG progress ring.

The target users for EcoTrace AI fall into four main groups:

Individual Households & Eco-Conscious Citizens

People who want to calculate their personal carbon footprint, identify their largest emission sources, and receive actionable tips to reduce their lifestyle impact.

Educators, Students, & Climate Researchers

Individuals looking for an educational tool to study SDG 13 (Climate Action). The visual neural network simulator provides a hands-on way to teach how machine learning algorithms function.

Green Technology & Smart-Home Enthusiasts

Homeowners interested in clean energy (solar panels, batteries, heat pumps) who want to understand how intelligent grids manage battery charge cycles to save energy and reduce grid stress.

Community Organizers & Environmental Groups

Groups that can use the gamified checklist and tracking metrics to run environmental challenges in schools, offices, or local neighborhoods.

Project Features

1. Interactive Carbon Footprint Calculator

Provides a multi-step wizard to input utility bills, vehicle travel, flights, and dietary preferences. It calculates your personal emissions in real-time, instantly rendering data breakdowns across interactive Chart.js doughnut and line graphs. (Located in

index.html

&

app.js

)

2. AI Recommendation Engine

Analyzes calculated parameters to isolate your highest emissions sectors. It automatically lists the top 3 weighted, personalized actions you can take to lower your carbon footprint, along with estimated savings. (Located in

ai-engine.js

)

3. "EcoBot" NLP Chat Assistant

A client-side conversational AI that responds to user inquiries about sustainability and green living. It uses keyword matching to provide detailed explanations on topics like carbon offsets, smart grids, and SDG 13. (Located in

ai-engine.js

)

4. Smart Grid Neural Network Simulator

Simulates a live Artificial Neural Network (ANN) training on a canvas to optimize home solar usage and battery reserves. It visualizes synapses, weights, and electrical activations, demonstrating how AI can balance energy load. (Located in app.js)

5. Eco Goal & Impact Tracker

A gamified checklist of sustainable commitments that deducts carbon offsets from your score when marked complete. It features an animating SVG radial progress circle and celebrates goal completion with confetti bursts.

Technologies used in app

1. Core Front-End Stack

HTML5: For semantic layout, form wizard structures, the chatbot window, the canvas visualizer, and SEO tags.

Vanilla CSS3: Used to design the interface with custom CSS variables, glassmorphic backdrop filters, custom scrollbars, responsive grids, and interactive keyframe animations.

Vanilla JavaScript (ES6+): Manages state, event listeners, calculations, and visual animations.

2. CDNs & Libraries

Chart.js: Renders interactive doughnut charts and line graphs showing footprint breakdowns and historical data trends.

Canvas Confetti: Handles celebration particle bursts when completing eco-commitments.

FontAwesome (v6): Supplies high-quality icons for tabs, calculators, and chatbot prompts.

Google Fonts: Uses Outfit for modern headings and Inter for clean, legible body text.

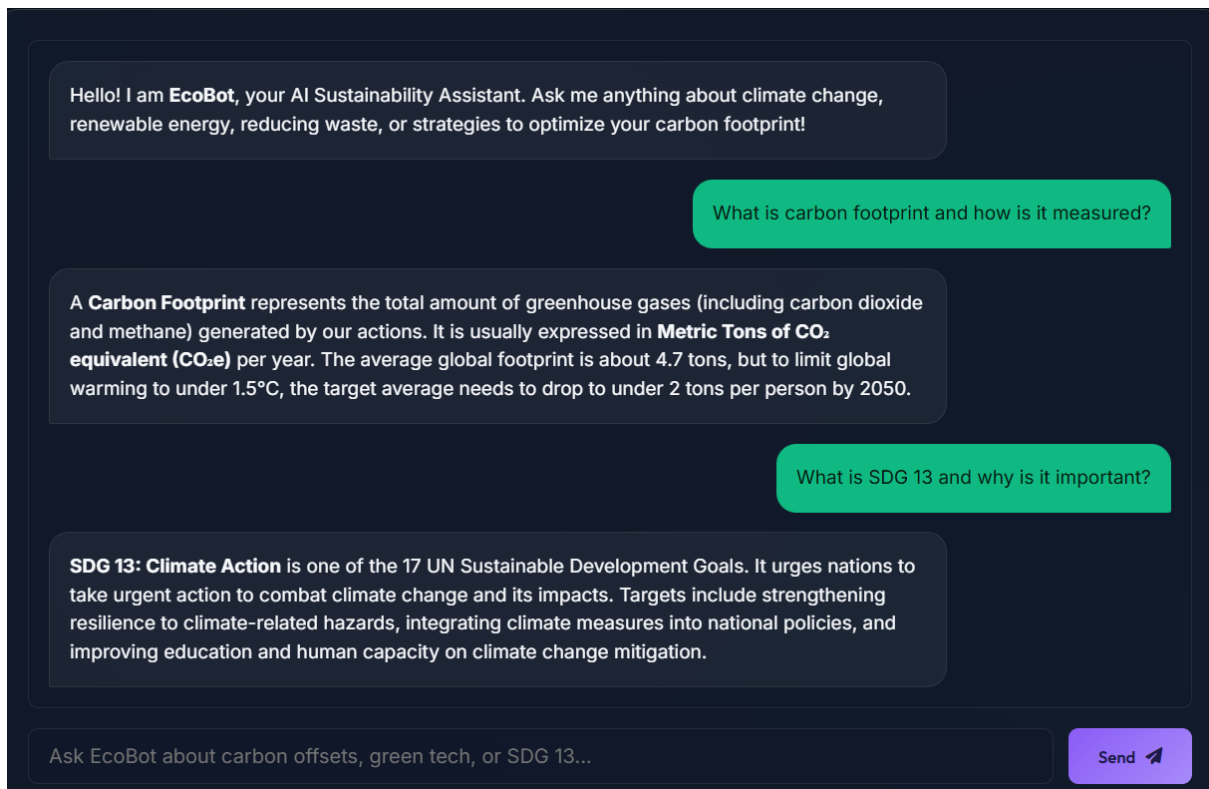
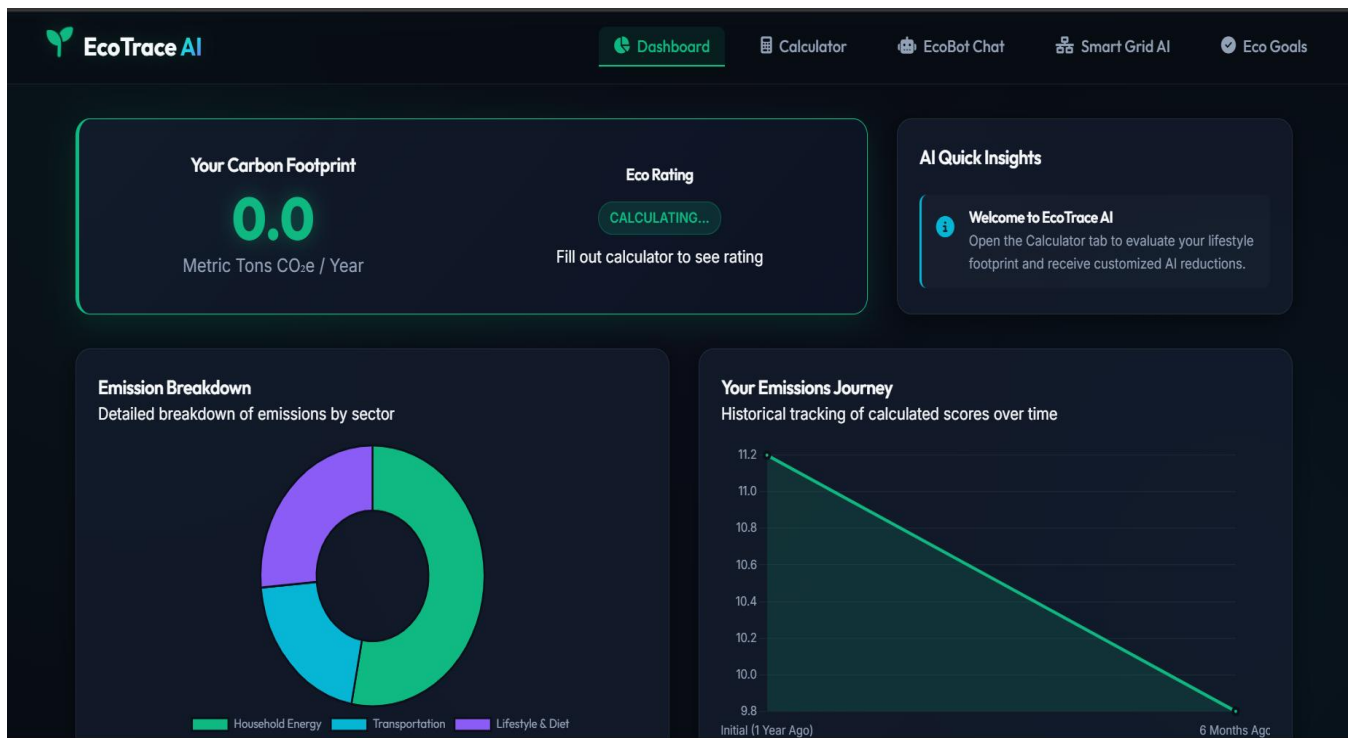
3. Client-Side AI Engines (Built from Scratch)

Custom Artificial Neural Network (ANN): Built with backpropagation and sigmoid activation models to run machine learning training directly in the browser.

Rule-Based Recommendation Engine: Processes input scoring matrices in ai-engine.js to output carbon-reduction tips.

NLP Token Matcher: Handles conversational responses by parsing user input keywords inside the chatbot.

Project Screenshots



Open source Repository

<https://github.com/shirvalkarsanskar123-pixel/sanskar>

Future Scope

IoT Smart Home Integration: Link the simulator to actual smart home APIs (like Home Assistant) to switch appliance loads based on real-time carbon grid signals.

Machine Learning API: Expand the rule-based recommender to consume multi-user datasets and utilize clustering algorithms (like K-Means) to predict user conservation success rates.

Global Database Syncing: Integrate a cloud database (such as Firebase) to enable users to log in, save historical footprint timelines permanently, and compete in community eco-leaderboards.